
ClimateKG: Context-Gated Scientific GraphRAG for Literature-Grounded Land-Use Scenario Design

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Abstract

1 Land-use and land-cover changes alter surface energy, water, and momentum ex-
2 changes, but the resulting cloud, rainfall, and temperature responses depend on
3 season, hydroclimate, background wind, and intervention geometry. Testing many
4 spatial land-use scenarios with convection-resolving models is expensive, while
5 ordinary retrieval can combine topically relevant but environmentally incompat-
6 ible findings. We introduce ClimateKG, a scientific GraphRAG system that ex-
7 tracts evidence-backed claims together with the complete study settings under
8 which they hold, gates evidence by a query’s environmental context before graph
9 traversal, and returns answers traceable to exact source passages. A staged run
10 completed 17 papers, producing 2,583 evidence blocks, 155 study contexts, and
11 261 scientific claims; a separate ten-paper end-to-end validation exercised graph
12 construction and cited querying. Two case studies show why conditioning mat-
13 ters: patch size and wind alter heterogeneous-surface convection, while the same
14 observed deforestation pattern yields null, cloud-only, or rainfall responses across
15 atmospheric regimes. The system is intended to propose auditable hypotheses and
16 a small set of candidate land-use rasters for subsequent WRF evaluation, not to
17 prescribe land management or replace simulation.

18 1 Why context-conditioned evidence?

19 Land surfaces influence temperature, boundary-layer growth, clouds, rainfall, and circulation
20 through albedo, roughness, soil moisture, and evapotranspiration [Pielke et al., 2007, Cao et al.,
21 2020]. These effects are not intervention lookup rules. Mesoscale soil-moisture patterns can or-
22 ganize storm initiation [Taylor et al., 2011]; changing patch size or background wind can change
23 whether and where convection develops [Lee et al., 2019]. Likewise, observed Amazonian deforesta-
24 tion produced different cloud and rainfall responses under different seasonal and synoptic regimes
25 [Wang et al., 2000]. A useful literature system must therefore retrieve not only *what* changed and
26 *what* followed, but *where*, *when*, *at what scale*, and *under which atmospheric state*.

27 The motivating application is scenario design. Exploring many high-resolution land-cover layouts
28 for many regions with a numerical model such as WRF [Skamarock et al., 2019] is computationally
29 demanding; multi-year convection-resolving regional simulations illustrate the scale of this cost
30 [Leutwyler et al., 2017]. Literature can narrow this search: retrieve transferable mechanisms, iden-
31 tify conditions and counter-evidence, generate a small set of spatially explicit candidate land-use
32 rasters, and then test those candidates numerically. Plain RAG [Lewis et al., 2020] provides passage

Indexing: PDF → Nemotron Parse v1.2 → cleaned **EvidenceBlocks** → hierarchical **StudyContexts** + **ContextFacets** → **LandTransitions** + evidence-backed **ScientificClaims** → canonical concepts + embeddings → Parquet/Neo4j graph.
Querying: question + optional local data → structured query context → context candidates and coverage → claim gate → compatible mechanism paths → smallest supporting EvidenceBlocks → cited answer.

Figure 1: ClimateKG makes the scientific setting part of retrieval. Claims are gated before they can participate in a mechanism path.

33 evidence, and generic GraphRAG supports corpus-level connections and summaries [Edge et al.,
 34 2024]; neither representation makes environmental applicability a first-class constraint.

35 We contribute (1) a provenance-preserving representation of scientific findings and their study set-
 36 tings; (2) context gating before mechanism-path search, while separating missing information from
 37 mismatch; (3) staged and single-call extraction routes for different paper lengths; and (4) a repro-
 38 ducible pilot with saved prompts, model responses, intermediate JSON, graph tables, query reports,
 39 and exact citations. The contribution is an evidence-to-hypothesis system, not evidence that any
 40 retrieved intervention will work locally.

41 2 ClimateKG

42 **Scientific representation.** An **EvidenceBlock** is a stable, page-linked passage from the current
 43 paper. A **StudyContext** is one complete observational or experimental setting; a hierarchy shares
 44 common setup while preserving children such as the same region in different seasons or the same
 45 simulation at different patch sizes and wind speeds. A **ContextFacet** records one relevant property
 46 (e.g., atmosphere, climate, hydrology, land surface, terrain, or spatial configuration). A **LandTran-**
 47 **sition** links baseline and changed settings. A **ScientificClaim** connects source and target states,
 48 marks causal versus associative language, and cites the smallest directly supporting blocks. These
 49 reader-facing names correspond to SourceBlock, Context, Facet, Transition, and Claim in
 50 the released schemas.

51 **Indexing.** NVIDIA Nemotron Parse v1.2 is the sole PDF parser. Deterministic cleaning removes
 52 repeated furniture and marks reference material for exclusion from semantic extraction while pre-
 53 serving sections, tables, captions, units, and provenance. The local Qwen3.6–27B model is used
 54 only for semantic extraction and reconciliation; Pydantic schemas reject malformed references
 55 rather than silently repairing scientific content. Long papers use a staged route: section scouting,
 56 a compact hierarchy of complete settings, facet extraction, claim extraction, consolidation, and state
 57 canonicalization. Short cleaned papers may use one structured extraction call followed by the same
 58 deterministic consolidation and graph stages. Qwen3-Embedding-4B produces normalized 2,048-
 59 dimensional vectors. Embedding similarity proposes candidates; it does not force uncertain concept
 60 merges.

61 **Context retrieval and claim gating.** The query parser returns optional source/target states, an in-
 62 tervention, and ContextFacets. For same-domain query facet q and candidate facet f , pair similarity
 63 is

$$s(q, f) = \alpha \cos(E_n(q), E_n(f)) + (1 - \alpha) \cos(E_c(q), E_c(f)), \quad (1)$$

64 where E_n embeds the domain and notion and E_c includes the description. Directional MaxSim
 65 averages each query facet’s best same-domain match. Coverage is reported separately, and the
 66 ranking score is $S_{ctx} = S_{sem}[1 - \lambda(1 - \text{coverage})]$; absent information is therefore not treated
 67 as contradiction. Approximate nearest-neighbor search proposes contexts, followed by exact facet-
 68 level reranking.

69 A claim first inherits applicability from its supporting StudyContext (or the baseline context of its
 70 transition). Only facets that materially condition that claim affect its conditioning score. Context,
 71 mechanism, intervention, and endpoint signals remain separate in the report and are combined only

Table 1: Measured artifacts from saved pilot runs. Times include cached parsing in the combined experiment, so route times are not a controlled end-to-end speed comparison.

	Staged run	Combined run
Registered / completed papers	20 / 17	10 / 7
EvidenceBlocks	2,583	906
StudyContexts / ContextFacets	155 / 497	46 / 60
LandTransitions / ScientificClaims	54 / 261	20 / 65
Mean successful paper time	20.02 min	3.29 min
LLM calls	305	10

for ranking. Gating precedes beam search, preventing a path from joining individually plausible claims supported by incompatible settings. Spatial relations retain distinct rules for local advection, patch/edge effects, distance decay, windward/leeward effects, moisture recycling, and teleconnections. A cardinal recommendation requires local geometry and transferable evidence.

Grounded answer. The synthesis prompt sees only selected claims, context scores, and their supporting EvidenceBlocks. Evidence-budget trimming cannot retain a claim after removing all its support. Every substantive answer statement links to a claim, paper, page, and block; scores are ranking signals, not probabilities that a finding is true.

3 Pilot validation

Execution and efficiency. We selected PDFs spanning heterogeneity, irrigation, vegetation change, deforestation, albedo, roughness, and soil-moisture feedbacks. Table 1 reports saved runs, not a claim of corpus representativeness. The staged run registered 20 papers and completed 17; three were visibly retained for review. It took 20.02 minutes per completed paper on average (20.92 median), with paper mapping, facet extraction, parsing, and claim extraction dominating time. The completed graph contains 6,057 relationships and 1,341 evidence links. In a separate ten-paper combined-route experiment, 7/10 papers passed all checks using one extraction call per paper; successful papers averaged 3.29 minutes. The three rejected outputs contained invalid evidence or conditioning references. This route is faster, but its 70% completion rate motivates routing and fail-closed validation rather than treating it as universally interchangeable with staged extraction.

End-to-end and mechanism cases. A separate ten-paper validation built 2,184 EvidenceBlocks, 101 StudyContexts, 437 ContextFacets, 51 LandTransitions, and 213 ScientificClaims, then exercised Parquet/Neo4j ingestion and ten fixed query workflows. All ten produced schema-valid cited answers after defect-focused reruns. This verifies pipeline execution and traceability, not answer correctness at benchmark scale.

Table 2 presents two manually checked cases. For the in-corpus heterogeneous-surface query, parsing preserved the 14.4-km chessboard and zero-wind facets. The top context had full facet coverage ($S_{ctx} = 0.676$), and the answer cited the finding that patches above 5 km transitioned from shallow cumulus to deep precipitating convection over dry patches, while homogeneous forcing retained shallow clouds. It also preserved the counterintuitive result that cross-patch circulation reduced the near-surface dry-wet temperature contrast [Lee et al., 2019]. With 1 m s^{-1} wind, a separate claim moves convergence from the dry-patch center to its downwind edge, so collapsing the cases would produce a misleading spatial answer.

The Rondonia extraction separates rainy, break, and dry-season simulations. Under rainy-season strong synoptic forcing, control and deforestation rainfall were nearly identical and mesoscale circulations were inhibited. During the break, weakened winds allowed enhanced low cloud but little colocated rainfall. In the dry-season setting, weak wind allowed a stronger land-induced circulation and localized convective response [Wang et al., 2000]. The same land pattern therefore cannot be mapped to one unconditional “deforestation increases rainfall” edge.

Table 2: Why complete study settings matter. IDs point to released claims and exact supporting blocks in the appendix.

Case	Preserved conditions	Indexed/queried conclusion
Heterogeneous LES	14.4-km dry/wet chessboard; zero wind	> 5-km patches support deep precipitating convection over dry patches; cross-patch mixing reduces surface contrast (CL002, CL004).
Rondonia deforestation	Rainy/break/dry season; original vs. weakened synoptic wind	Null rainfall under strong forcing; cloud-only response in weakened-wind break case; dry-season circulation response (CL001, CL006–010, CL015).
New Rajasthan location	Semi-arid; 10–15-km alternating irrigated/dry patches; weak monsoon wind	Literature analogy can motivate a mechanism hypothesis, but cannot establish local rainfall or cardinal placement without local geometry and numerical/field validation.

Frozen query audit. We froze three questions before inspection: the in-corpus 14.4-km zero-wind LES case; the in-corpus Rondonia seasonal/wind comparison; and an out-of-corpus semi-arid Rajasthan case with weak monsoon wind and proposed 10–15-km irrigated/dry patches. For each answer, the saved audit records direct support, citation coverage, conditioning variables, incompatible evidence, causal wording, unsupported local advice, latency, and tokens. The first query passed manual trace inspection. The second revealed the downstream recall failure described below. The third returned cited cloud/edge mechanisms and an explicit transfer warning in 4.84 minutes, but omitted the requested patch scale and did not reconcile weak user-specified wind with derived JJA wind of 3.09 m s^{-1} . These cases are diagnostic, not a powered benchmark.

The packaged plain RAG and Microsoft GraphRAG implementations enable a later equal-corpus comparison. A valid comparison must hold the questions, generation model, evidence budget, and source corpus fixed; separately score applicable-claim retrieval, incompatible-evidence rate, citation support, and answer completeness; and report latency and compute. Community summarization and context-gated path retrieval serve different purposes, so we do not interpret one case-study answer as universal system superiority. A held-out expert annotation set is still required.

4 Climate impact pathway and limits

For scientists, ClimateKG exposes mechanism paths and contradictions for literature synthesis. For regional planners, it can translate a local question into a shortlist of evidence-backed scenarios and explicit unknowns. For farmers or communities, it may support transparent discussion, but should not issue autonomous recommendations. A proposed workflow is: define the target region and objective; derive citable local climate, terrain, land-cover, wind, and geometry variables; retrieve compatible findings; generate a small, diverse set of LULC rasters; evaluate them with WRF or another appropriate model; and assess water, food, biodiversity, livelihood, and justice trade-offs with local experts and affected communities.

The present corpus is small and geographically biased; extraction still fails; similarity is not causal validity; and no WRF experiment, field validation, stakeholder study, or statistical system-superiority benchmark has been completed. A frozen 17-paper Rondonia query also exposed a downstream failure: the correct paper’s context ranked sixth and its claims entered the candidate set, but source-state mapping was cleared and path search retained only generic deforestation paths. The answer appropriately reported missing seasonal evidence, yet failed to use available evidence. This localizes an important limitation to path seeding/selection rather than extraction. The new-location case is consequently a transferability audit, not a recommendation. Plain RAG and Microsoft GraphRAG are packaged for controlled comparison, but we do not report unfinished baseline results. Immediate priorities are expert annotation, citation/path audits, equal-budget baselines, and numerical testing of selected raster scenarios. Even before those stages, the pilot shows that complete study settings prevent distinct regimes from being flattened into one literature edge and keep generated statements inspectable at source.

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177 A Reproducibility and extraction detail

178 A.1 Released artifacts

179 The repository preserves the rendered/cleaned document, stable EvidenceBlocks, every completed
180 LLM request and response, schema-validation failures, consolidated paper JSON, embeddings, Par-
181 quet graph tables and ANN indexes, and per-query reports. The two extraction routes converge on
182 the same final schema. The staged route is:

```
183         section scout → hierarchy of complete experimental settings → ContextFacet extraction  
184         → ScientificClaim extraction → context reconciliation → paper consolidation → state  
185         canonicalization → embeddings.
```

186 The combined route supplies all cleaned non-reference EvidenceBlocks to a single structured ex-
187 traction prompt, then uses the same reconciliation, consolidation, state, embedding, and graph steps.
188 The routing count uses the o200k_base tokenizer on cleaned non-reference blocks; 8,000 tokens is
189 a preferred target rather than a hard rejection boundary, with a configured 16,000-token safety limit
190 for the experiment reported here.

191 A.2 Central object schemas

192 Below, evidence_block_ids must name directly supporting passages from the current paper. IDs
193 are permanent after assignment.

```
194 StudyContext {  
195     id, paper_id, name, description,  
196     parent_context_id?, evidence_block_ids[], context_embedding  
197 }  
198 ContextFacet {  
199     id, paper_id, context_id,  
200     domain: spatial_configuration | land_surface | hydrology |  
201           atmosphere | climate | substrate_terrain | other,  
202     notion, description, origin, source?, evidence_block_ids[], embeddings  
203 }  
204 LandTransition {  
205     id, paper_id, from_context_id, to_context_id,  
206     intervention_description, evidence_block_ids[]  
207 }  
208 ScientificClaim {  
209     id, paper_id,  
210     scope_type: context | transition, scope_id,  
211     from: {concept, state}, to: {concept, state},  
212     relation: causal | associative,  
213     description, evidence_role,  
214     conditioning_facet_ids[], evidence_block_ids[], claim_embedding  
215 }
```

216 B Staged trace: heterogeneous surface experiment

217 The indexed paper is Lee et al. [Lee et al., 2019], DOI 10.1175/JAS-D-18-0196.1. Nemotron
218 Parse produced 163 EvidenceBlocks. Staged extraction produced 26 StudyContexts, 59 Con-
219 textFacets, four LandTransitions, and 18 ScientificClaims. The setting hierarchy retains the com-
220 mon LES setup and separates heterogeneous patch size/wind combinations from homogeneous dry,
221 wet, and domain-mean controls. For example, the 14.4-km case has distinct zero, 1, and 2 m s⁻¹
222 children rather than one context with wind described in prose.

Claim CL002. P000001_CL002 is a causal claim scoped to transition P000001_T001; it is conditioned by facet P000001_F012. Its description states that chessboard heat-flux heterogeneity with patches above 5 km produces a transition from shallow cumulus to deep precipitating convection over dry patches, whereas homogeneous domain-mean fluxes retain scattered shallow clouds. The sole evidence is P000001:S06:P0033, which reports deep clouds above 6 km and significant precipitation over dry patches for large zero-wind patches, versus shallow cumulus for patches below 5 km.

Claim CL004. P000001_CL004 uses P000001:S11:P0112 to record that active cross-patch circulation horizontally mixes the boundary layer and reduces the near-surface dry–wet temperature contrast. This is easy to omit from a simple “heterogeneity causes convection” summary, yet it is important because the surface-temperature response and cloud response do not follow the same naive contrast argument.

Wind-dependent spatial claim. P000001_CL014 cites P000001:S11:P0097 and P000001:S12:P0122: changing background wind from zero to 1 m s^{-1} moves the convergence/updraft from the dry-patch center to the downwind edge; sufficiently stronger flow destroys the organized circulation. This claim belongs to a different transition and cannot be substituted into the zero-wind answer.

B.1 Query trace

The frozen question was: “Under zero background wind over a heterogeneous chessboard surface with 14.4 km dry and wet patches, how does surface heat-flux heterogeneity affect local cloud development and precipitation?” The parser created two user-origin facets: (i) spatial configuration, notion *chessboard surface pattern*, description *14.4 km patches*; and (ii) atmosphere, notion *zero background wind*. The top retrieved context matched both domains (coverage 1.0), with spatial and atmospheric pair scores 0.662 and 0.691 and overall context score 0.676. After claim gating, synthesis cited CL002 for deep precipitating convection, CL001 for stronger updrafts, CL003 for mixed-layer thermodynamics, and CL004 for reduced surface contrast. The response explicitly limited transfer to the described patch configuration and did not invent a local cardinal placement.

C Staged trace: Rondonia deforestation

The indexed study is Wang et al. [Wang et al., 2000]. Its 142 EvidenceBlocks yielded 20 StudyContexts, 44 ContextFacets, eight LandTransitions, and 33 ScientificClaims. The hierarchy separates break-period original wind, break-period weakened wind, rainy-season strong forcing, and dry-season weak-wind settings.

Table 3: Selected extracted findings and their exact evidence anchors.

Claim	Extracted result	EvidenceBlock(s)
CL001	Break period, original wind: at most 0.1 mm at isolated control locations and almost no rainfall in the deforestation run.	S08:P0056
CL006–007	Weakened wind: enhanced low cumulus at 900–800 hPa over deforested areas, but sparse rain cells did not overlap the enhanced cloud.	S08:P0062, P0064
CL008–010	Rainy season: control and deforestation rainfall/cloud distributions were nearly identical because stronger synoptic forcing inhibited mesoscale circulation.	S09:P0072, P0074; S11:P0101
CL015	Dry season, weak wind: significant low-level mesoscale heat flux indicated a land-induced circulation; related evidence reports localized convection/rainfall on the first day.	S10:P0091, P0092

255 This trace illustrates the representation’s purpose. A graph with one unconditional *deforestation* →
256 *rainfall* edge would erase null and cloud-only regimes and could reverse the interpretation of a local
257 query.

258 C.1 Observed retrieval failure

259 We ran the frozen Rondonia question over the 17-paper Parquet graph. The parser correctly cre-
260 ated rainy/break/dry-season and synoptic-wind facets and added provenance-bearing local enrich-
261 ment. The best P000019 StudyContext ranked sixth (context score 0.495); P000019 claims includ-
262 ing CL001, CL007, CL008, CL010, and CL015 appeared among claim candidates. Nevertheless,
263 the state-mapping stage emitted QUERY_UNSUPPORTED_SOURCE_STATE_CLEARED, and all four re-
264 tained mechanism paths used generic P000002 deforestation claims. Synthesis therefore stated that
265 the supplied evidence lacked seasonal and wind-conditioned findings. This was a grounded absten-
266 tion relative to its selected evidence, but an end-to-end retrieval error because applicable indexed
267 evidence existed. The saved report localizes the defect to source-state/path seeding and evidence
268 selection after context retrieval.

269 D New-location transfer and baselines

270 The frozen out-of-corpus question describes a semi-arid Rajasthan planning area, strong daytime
271 heating, weak monsoon background wind, and proposed alternating irrigated/dry patches of 10–15
272 km. A place name alone is not used to infer climate. Query enrichment may add reproducibly
273 sourced local facets (coordinates/geometry, climate class, aridity, elevation, wind, land cover, and
274 patch metrics), with each derived value retaining dataset and operation provenance. Retrieval may
275 then identify analogous patch-circulation mechanisms, but the answer must report missing or mis-
276 matched facets and cannot assert local rainfall, optimal LULC, or cardinal placement without appli-
277 cable evidence and local numerical or field validation.

278 **Observed out-of-corpus result.** The query resolved Rajasthan to an approximate point and de-
279 rived a TerraClimate aridity ratio of 0.209 (semi-arid), BWh Köppen–Geiger class, cropland land
280 cover, terrain, and seasonal ERA5-Land winds with dataset provenance. End-to-end latency was
281 290.4 s: query parsing used 12.9 s (882 prompt and 325 generated tokens), while synthesis used
282 232.3 s (19,224 prompt and 1,026 generated tokens). The answer cited mechanisms for enhanced
283 afternoon cloud over vegetated/cropland surfaces and downwind edge effects, and explicitly warned
284 that the evidence came from other climate regimes and did not establish transfer to Rajasthan. How-
285 ever, the parser encoded the 10–15-km alternating pattern in the source state rather than a spatial-
286 configuration facet, and the answer omitted that scale. It also failed to reconcile the user’s weak-
287 monsoon description with derived JJA wind of 3.09 m s^{-1} . The run therefore passed basic ground-
288 ing/abstention checks but failed the full conditioning audit.

289 The release includes three runnable systems: the proposed indexing/query pipeline, a project-owned
290 plain RAG baseline, and Microsoft GraphRAG 3.1.1 under its MIT license. A frozen comparison file
291 specifies the same three questions, local Qwen3.6–27B generator, shared embedding model where
292 supported, corpus, and audit fields. Required audits are direct support, citation precision/coverage,
293 preservation of conditioning variables, incompatible evidence use, causal/associative language, un-
294 supported location-specific recommendations, latency, and token use. We deliberately omit incom-
295 plete baseline scores from the main paper.

296 E Known limitations and next validation steps

297 The pilot is not a representative literature census. PDF errors, extraction omissions, schema failures,
298 canonicalization mistakes, and incomplete local enrichment remain possible. A claim’s retrieval
299 score is not scientific confidence, external validity, feasibility, or net climate benefit. Before plan-
300 ning use, the system requires a held-out, expert-annotated query benchmark; independent citation

301 and path review; geographic coverage analysis; equal-budget RAG/GraphRAG baselines; sensitiv-
302 ity analysis for context weights and thresholds; and WRF experiments for selected scenarios. Any
303 intervention also requires water, biodiversity, food, livelihood, governance, and equity assessment
304 with affected communities.